Model Architecture: Qwen3.5-0.8B

Generated by sg_modelload

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1 Model Overview

Model name	Qwen3.5-0.8B
Transformers class	`Qwen3_5ForConditionalGeneration`
Path	`/models/memorization/Qwen3.5-0.8B`
Total parameters	852,985,920 (852.99M)
Checkpoint-only tensors (not in total)	20,452,864
Weight dtype	bfloat16
Hidden size d	1,024
Decoder layers L	24
Query heads H	8
Key/value heads H_{kv}	2
Head dim d_h	256
MLP width d_{ff}	3,584
DeltaNet q/k heads	16
DeltaNet v heads	16
DeltaNet q/k head dim	128
DeltaNet v head dim	128
DeltaNet conv kernel	4
Vocabulary V	248,320
Max context length	262,144
RMSNorm ϵ	1e-06
Tied input/output embeddings	True
Activation	SiLU
RoPE base θ	10,000,000
Vision encoder: layers	12
hidden size / heads	768 / 12

How to read this document. Section 2 treats the model mathematically, top-down: first the whole network as one composition $\Psi \circ \mathcal{B}^{(L)} \circ \cdots \circ \mathcal{B}^{(1)} \circ E$ of layers, then a definition of every layer down to single neurons and attention heads, then parameters, outputs, intermediate states and the local linear response. Section 3 gives readable pseudocode for the forward pass and for every distinct block type. Section 4 lists the weights of each block type *once*, with how many times it repeats. Section 5 explains in plain language what every kind of layer does. The original layer-by-layer table with every repetition written out is in Appendix A.

Basic notation

B, T	batch size, sequence length (tokens)
d	hidden size (1024)
V	vocabulary size (248,320)
H, H_{kv}, d_h	query heads, key/value heads, size of one head
W_*, b_*	learned weight matrix / bias of the named projection
$\odot$, σ	element-wise product, logistic sigmoid
$f^{\circ n}$	f applied n times in a row ($f \circ f \circ \cdots \circ f$)
$g \circ f$	first f, then g (composition reads right to left)
RMSNorm(x)	$\gamma \odot x / \sqrt{\text{mean}(x^2) + \epsilon}$ (learned scale γ)
LN(x)	LayerNorm: $\gamma \odot (x - \mu)/\sigma + \beta$
RoPE	rotary position embedding (rotates query/key pairs by an angle set by position)

2 Mathematical Representation

We describe the model *top-down*. First the whole network is written as one composition of functions (Section 2.1). Then every function that appears in that composition is defined (Section 2.2), down to individual neurons and attention heads. The remaining subsections describe the parameters, the outputs, the intermediate states, and the local linear behaviour.

2.1 The model as a composition of layers

Definition 2.1 (Qwen3.5-0.8B as an L-layer decoder)**.** The model is a function $f_\theta : \mathcal{X} \to \mathcal{Y}$ with parameters θ. It maps a sequence of T token ids $x \in \mathcal{X} = \{1, \ldots, V\}^T$ to one logit vector per position, $y \in \mathcal{Y} = \mathbb{R}^{T \times V}$

($V = 248{,}320$). It is a composition of an embedding E_{mm}, $L = 24$ decoder blocks $\mathcal{B}^{(\ell)}$ and an output head Ψ:

$$f_\theta \;=\; \Psi \circ \mathcal{B}^{(L)} \circ \mathcal{B}^{(L-1)} \circ \cdots \circ \mathcal{B}^{(2)} \circ \mathcal{B}^{(1)} \circ E_{\text{mm}} \tag{1}$$

Every block maps $\mathbb{R}^{T\times d} \to \mathbb{R}^{T\times d}$ ($d = 1024$), so the blocks can be stacked in any number. Block ℓ has its own parameters θ_ℓ and is stored in the checkpoint as `model.language_model.layers.`$(\ell - 1)$.

Definition 2.2 (Block types)**.** There are 2 block types (same function form within a type, different weights). A type map $c : \{1, \ldots, L\} \to \{\text{L}, \text{F}\}$ says which type layer ℓ is: $\mathcal{B}^{(\ell)} = \mathcal{B}_{c(\ell)}(\,\cdot\,; \theta_\ell)$. Here $c = ((\text{L}^3\,\text{F})^6)$, read left to right from layer 1. Written out, with the type as a subscript:

$$f_\theta = \Psi \circ \mathcal{B}_{\text{F}}^{(24)} \circ \cdots \circ \mathcal{B}_{\text{F}}^{(4)} \circ \mathcal{B}_{\text{L}}^{(3)} \circ \mathcal{B}_{\text{L}}^{(2)} \circ \mathcal{B}_{\text{L}}^{(1)} \circ E_{\text{mm}} \tag{2}$$

Grouping the repetitions with composition powers ($\mathcal{B}^{\circ n} = n$ consecutive blocks of the same type, each with its own weights) gives the compact form

$$f_\theta = \Psi \circ \Big(\big(\mathcal{B}_{\text{F}} \circ \mathcal{B}_{\text{L}}^{\circ 3} \big)^{\circ 6} \Big) \circ E_{\text{mm}} \tag{3}$$

Definition 2.3 (Vision encoder)**.** The vision encoder is itself a composition of 12 blocks acting on a sequence of image patches of width 768:

$$\text{Enc}_{\text{V}} = \text{Post} \circ \mathcal{B}_{\text{V}}^{\circ 12} \circ \text{PatchEmbed} \tag{4}$$

PatchEmbed turns the raw input into vectors and Post is the final norm / pooling / merging (Table 3).

Layer pattern.

Symbol	Block type	Count	Params / block	Checkpoint indices
$\mathcal{B}_{\text{L}}$	Linear-attention (Gated DeltaNet) block	18	21,555,360	0–2, 4–6, 8–10, 12–14, 16–18, 20–22
$\mathcal{B}_{\text{F}}$	Full (global) attention block	6	18,352,640	3, 7, …, 23 (every 4th)
$\mathcal{B}_{\text{V}}$	Vision encoder block	12	7,087,872	0–11

Language-model decoder (24 layers). layers 0–23: [$\mathcal{B}_{\text{L}}\times 3$, $\mathcal{B}_{\text{F}}$] repeated 6× (checkpoint indices, i.e. $\ell - 1$).

Vision encoder (12 layers). layers 0–11: [$\mathcal{B}_{\text{V}}$] repeated 12× (checkpoint indices, i.e. $\ell - 1$).

2.2 The layers, top to bottom

Every function in the composition above is defined here. Only the attention (or linear-attention) sub-layer mixes information *between* positions t. Every other sub-layer (norms, feed-forward networks, projections) acts on each token vector $h_t \in \mathbb{R}^d$ independently, e.g. $\text{FFN}(h)_t = \text{FFN}(h_t)$. All weights follow the convention $W \in \mathbb{R}^{d_{\text{out}}\times d_{\text{in}}}$, so a projection of a token vector is Wu (as in an affine layer $f_i(u) = A^{(i)}u + b^{(i)}$).

Definition 2.4 (Token embedding)**.** $E : \{1, \ldots, V\}^T \to \mathbb{R}^{T\times d}$ looks up one row of the embedding matrix $W_E \in \mathbb{R}^{V\times d}$ per token:

$$E(x)_t = W_E[x_t, :], \qquad t = 1, \ldots, T. \tag{5}$$

With images, the multimodal embedding replaces the placeholder-token rows of $E(x)$ by projected encoder outputs:

$$E_{\text{mm}}(x, \cdot) = \text{Merge}\Big(E(x),\ \text{Proj}_{\text{img}}\big(\,\text{Enc}_{\text{V}}(\text{img})\big)\Big). \tag{6}$$

Definition 2.5 (Normalisation)**.** $\text{RMSNorm}(u) = \gamma \odot u / \sqrt{\frac{1}{d}\sum_i u_i^2 + \epsilon}$, with a learned scale $\gamma \in \mathbb{R}^d$, applied to each token vector separately. It fixes the scale of a vector without changing its direction. Every norm in the model has its own γ, distinguished by a subscript (e.g. RMSNorm_1, $\text{RMSNorm}_{\text{final}}$). LN (LayerNorm, used in some encoder blocks) additionally subtracts the mean and adds a bias.

Definition 2.6 (Block $\mathcal{B}_{\text{L}}$: Linear-attention (Gated DeltaNet) block)**.** A decoder block of this type maps $h \in \mathbb{R}^{T\times 1024}$ to the same space. It is a composition of 2 residual sub-layers, each of the form $(\text{id} + g)(h) = h + g(h)$:

$$\mathcal{B}_{\text{L}} = (\text{id} + \text{FFN} \circ \text{RMSNorm}_2) \circ (\text{id} + \text{Mixer} \circ \text{RMSNorm}_1) \tag{7}$$

Step by step, the sub-layers are:

$$a = h + \text{Mixer}(\text{RMSNorm}_1(h)) \qquad \text{pre-norm + residual}$$

$$
\begin{aligned}
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a)) && \text{pre-norm + residual}\\
[q,\, k,\, v] &= \mathrm{SiLU}\big(\,\mathrm{CausalConv1d}(W_{qkv}\,u)\big) && \text{short causal conv (kernel 4) over time}\\
z &= W_z\,u && \text{output gate}\\
\beta_t &= \sigma(W_b\,u_t) && \text{write strength in (0,1)}\\
g_t &= -e^{A_{\log}} \cdot \mathrm{softplus}(W_a\,u_t + b_{dt}) && \text{decay } (\le 0)\text{: how fast memory fades}\\
q_t,\, k_t &= \mathrm{L2Norm}(q_t),\ \mathrm{L2Norm}(k_t)\\
S_t &= e^{g_t}\,S_{t-1}\big(I - \beta_t k_t k_t^\top\big) + \beta_t\,v_t k_t^\top && \text{fixed-size memory per head (delta rule)}\\
o_t &= S_t\,q_t && \text{read memory with the query}\\
\mathrm{Mixer}(u) &= W_{\mathrm{out}}\big(\,\mathrm{RMSNorm}(o)\odot\mathrm{SiLU}(z)\big)\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\,\mathrm{SiLU}(W_{\mathrm{gate}}\,u)\odot W_{\mathrm{up}}\,u\big) && \text{gated MLP, width 3584}
\end{aligned}
$$

Parameters of one block ($|\theta_\ell|$ = 21,555,360): $K_{\mathrm{conv}} \in \mathbb{R}^{6144\times 1\times 4}$, $W_{\mathrm{out}} \in \mathbb{R}^{1024\times 2048}$, $W_{qkv} \in \mathbb{R}^{6144\times 1024}$, $W_z \in \mathbb{R}^{2048\times 1024}$, $W_b \in \mathbb{R}^{16\times 1024}$, $W_a \in \mathbb{R}^{16\times 1024}$, $W_{\mathrm{gate}} \in \mathbb{R}^{3584\times 1024}$, $W_{\mathrm{up}} \in \mathbb{R}^{3584\times 1024}$, $W_{\mathrm{down}} \in \mathbb{R}^{1024\times 3584}$, plus 2,208 parameters in norm scales, biases and other vectors. This type is used 18 times.

Definition 2.7 (Block $\mathcal{B}_{\mathrm{F}}$: Full (global) attention block)**.** A decoder block of this type maps $h \in \mathbb{R}^{T\times 1024}$ to the same space. It is a composition of 2 residual sub-layers, each of the form $(\mathrm{id}+g)(h) = h + g(h)$:

$$\mathcal{B}_{\mathrm{F}} = (\mathrm{id} + \mathrm{FFN}\circ\mathrm{RMSNorm}_2)\circ(\mathrm{id} + \mathrm{Attn}\circ\mathrm{RMSNorm}_1) \tag{8}$$

Step by step, the sub-layers are:

$$
\begin{aligned}
a &= h + \mathrm{Attn}(\mathrm{RMSNorm}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{RMSNorm}_2(a)) && \text{pre-norm + residual}\\
[Q,\, G] &= W_Q\,u && \text{query and an output gate}\\
K,\ V &= W_K\,u,\quad W_V\,u && H_{kv} = 2 \text{ key/value heads}\\
Q,\ K &= \mathrm{RMSNorm}_q(Q),\ \mathrm{RMSNorm}_k(K) && \text{per-head normalisation}\\
Q,\ K &= \mathrm{RoPE}(Q),\ \mathrm{RoPE}(K) && \text{rotate by position}\\
S_h &= s\,Q_h K_{g(h)}^\top + M && \text{mask } M\text{: causal}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\,V_{g(h)}\\
\mathrm{Attn}(u) &= W_O\big([O_1;\ldots;O_H]\odot\sigma(G)\big)\\
\mathrm{FFN}(u) &= W_{\mathrm{down}}\big(\,\mathrm{SiLU}(W_{\mathrm{gate}}\,u)\odot W_{\mathrm{up}}\,u\big) && \text{gated MLP, width 3584}
\end{aligned}
$$

Parameters of one block ($|\theta_\ell|$ = 18,352,640): $W_Q \in \mathbb{R}^{4096\times 1024}$, $W_K \in \mathbb{R}^{512\times 1024}$, $W_V \in \mathbb{R}^{512\times 1024}$, $W_O \in \mathbb{R}^{1024\times 2048}$, $W_{\mathrm{gate}} \in \mathbb{R}^{3584\times 1024}$, $W_{\mathrm{up}} \in \mathbb{R}^{3584\times 1024}$, $W_{\mathrm{down}} \in \mathbb{R}^{1024\times 3584}$, plus 2,560 parameters in norm scales, biases and other vectors. This type is used 6 times.

Definition 2.8 (Block $\mathcal{B}_{\mathrm{V}}$: Vision encoder block)**.** A vision-encoder block of this type maps $h \in \mathbb{R}^{T\times 768}$ to the same space. It is a composition of 2 residual sub-layers, each of the form $(\mathrm{id}+g)(h) = h + g(h)$:

$$\mathcal{B}_{\mathrm{V}} = (\mathrm{id} + \mathrm{FFN}\circ\mathrm{LN}_2)\circ(\mathrm{id} + \mathrm{Attn}\circ\mathrm{LN}_1) \tag{9}$$

Step by step, the sub-layers are:

$$
\begin{aligned}
a &= h + \mathrm{Attn}(\mathrm{LN}_1(h)) && \text{pre-norm + residual}\\
\mathcal{B}(h) &= a + \mathrm{FFN}(\mathrm{LN}_2(a)) && \text{pre-norm + residual}\\
[Q,\, K,\, V] &= W_{qkv}\,u + b_{qkv} && \text{one fused projection}\\
Q,\ K &= \mathrm{RoPE}_{2D}(Q),\ \mathrm{RoPE}_{2D}(K) && \text{rotate by (row, column) patch position}\\
S_h &= s\,Q_h K_h^\top + M && \text{mask } M\text{: bidirectional}\\
P_h &= \mathrm{softmax}(S_h) && \text{attention weights}\\
O_h &= P_h\,V_h\\
\mathrm{Attn}(u) &= W_O\,[O_1;\ldots;O_H] + b_O\\
\mathrm{FFN}(u) &= W_2\ \mathrm{GELU}_{\tanh}(W_1 u + b_1) + b_2 && \text{MLP, width 3072}
\end{aligned}
$$

Parameters of one block ($|\theta_\ell|$ = 7,087,872): $W_{qkv} \in \mathbb{R}^{2304\times 768}$, $W_O \in \mathbb{R}^{768\times 768}$, $W_1 \in \mathbb{R}^{3072\times 768}$, $W_2 \in \mathbb{R}^{768\times 3072}$, plus 9,984 parameters in norm scales, biases and other vectors. This type is used 12 times.

Definition 2.9 (Output head)**.** $\Psi : \mathbb{R}^{T\times d} \to \mathbb{R}^{T\times V}$ normalises the last hidden state and scores every vocabulary token, re-using the embedding matrix (tied weights):

$$\Psi(h)_t = W_E^\top\ \mathrm{RMSNorm}_{\mathrm{final}}(h)_t \tag{10}$$

2.3 Neurons and attention heads

Definition 2.10 (Neuron, pre-activation value and state)**.** The feed-forward sub-layer of block ℓ contains $d_{ff}^{(\ell)} = 3584$ neurons. Neuron j of layer ℓ is the scalar function of a normalised token vector $u \in \mathbb{R}^d$

$$\eta_j^{(\ell)}(u) = \mathrm{SiLU}\left(W_{\mathrm{gate},j}^{(\ell)} \cdot u\right)\left(W_{\mathrm{up},j}^{(\ell)} \cdot u\right), \tag{11}$$

where $W_{\mathrm{gate},j}$ is the j-th row of W_{gate}. The feed-forward output is the sum of the neuron outputs, each written along its own column of W_{down}:

$$\mathrm{FFN}_\ell(u) = \sum_{j=1}^{d_{ff}^{(\ell)}} \eta_j^{(\ell)}(u)\, W_{\mathrm{down}}^{(\ell)}[:, j]. \tag{12}$$

Across its L blocks the network has $N = \sum_{\ell=1}^{L} d_{ff}^{(\ell)} = 86{,}016$ feed-forward neurons. For input x at position t, with $u_t^{(\ell)}(x)$ the normalised input of the feed-forward sub-layer:

- $\mathcal{V}(\eta; x, t) = W_{\mathrm{gate},j}^{(\ell)} \cdot u_t^{(\ell)}(x) \in \mathbb{R}$ is the **pre-activation value** of neuron η (the argument of the non-linearity).
- $\mathcal{S}(\eta; x, t)$ is its **activation state**: *active* if $\mathcal{V} > 0$ and *inactive* if $\mathcal{V} \le 0$. The non-linearity is smooth, unlike ReLU, so an inactive neuron's output is strongly suppressed rather than exactly zero.

Definition 2.11 (Attention head and attention pattern)**.** Head h of an attention layer ℓ reads from the whole sequence through its attention pattern $P_h^{(\ell)}(x) \in \mathbb{R}^{T \times T}$ (row t = how much position t attends to each earlier position) and writes through its own $d \times d_h$ slice $W_{O,h}^{(\ell)}$ of W_O:

$$\mathrm{Attn}_\ell(u) = \sum_{h=1}^{H} \left(P_h^{(\ell)} V_{g(h)} \odot \sigma(G_h)\right) W_{O,h}^{(\ell)\top}. \tag{13}$$

$g(h)$ is the key/value head shared by query head h (grouped-query attention). The pattern P_h plays the role that the activation state plays for neurons: once P_h is fixed, the head is linear in its values. The network has 6 attention layers $\times$ $H = 8$ heads $= 48$ attention heads.

2.4 Architecture and parameters

Definition 2.12 (Architecture and parameters)**.** The **architecture** is the tuple of sizes and the layer-type sequence:

$$\mathcal{A} = (V, d, L, c, H, H_{kv}, d_h, d_{ff}) = (248{,}320,\ 1024,\ 24,\ ((\mathrm{L}^3\,\mathrm{F})^6),\ 8,\ 2,\ 256,\ 3584). \tag{14}$$

The **parameters** are $\theta = \{W_E,\ \theta_1, \ldots, \theta_L,\ \gamma_{\mathrm{final}},\ \theta_{\mathrm{Enc_V}},\ \theta_{\mathrm{Proj}}\}$ (W_E is shared with the output head). Grouping the blocks by type,

$$\begin{aligned} |\theta| &= \sum_{\text{types } c} n_c\, |\theta_c| + |\theta_{\mathrm{rest}}| \\ &= 18 \times 21{,}555{,}360 + 6 \times 18{,}352{,}640 + 12 \times 7{,}087{,}872 \\ &\quad + 269{,}819{,}136 \\ &= 852{,}985{,}920, \end{aligned}$$

where n_c is how often type c occurs and θ_{rest} holds everything outside the repeated stacks (embedding, final norm, encoder stems and projector, ...; Table 3).

2.5 Output representations

Definition 2.13 (Output representations)**.** For an input sequence x, the output can be read at three levels:

1. **Logits:** $y = f_\theta(x) \in \mathbb{R}^{T \times V}$, one score per vocabulary token at every position.
2. **Next-token distribution:** $p(x)_{t,k} = \dfrac{\exp(y_{t,k})}{\sum_{j=1}^{V} \exp(y_{t,j})} = \Pr\left(x_{t+1} = k \mid x_1, \ldots, x_t\right)$.
3. **Predicted next token:** $z(x) = \mathrm{argmax}_{k \in \{1,\ldots,V\}}\, y_{T,k}$ (greedy decoding). Generation appends $z(x)$ (or a sample from $p(x)_T$) to x and applies f_θ again; in practice the keys and values of earlier positions are cached.

The model is **causal**: y_t depends only on $x_1, \ldots, x_t$, because every attention mask hides later positions. Training minimises the cross-entropy $-\sum_t \log p(x)_{t, x_{t+1}}$.

2.6 Intermediate states and sub-network decomposition

To study what happens at one layer $\ell \in \{1, \dots, L\}$, split the composition into a prefix, the layer itself and a suffix:

$$f_\theta = \underbrace{\Psi \circ \mathcal{B}^{(L)} \circ \cdots \circ \mathcal{B}^{(\ell+1)}}_{G_{\ell+1}\ \text{(suffix)}} \circ \underbrace{\mathcal{B}^{(\ell)}}_{\text{layer } \ell} \circ \underbrace{\mathcal{B}^{(\ell-1)} \circ \cdots \circ \mathcal{B}^{(1)} \circ E_{\text{mm}}}_{F_{\ell-1}\ \text{(prefix)}} \tag{15}$$

- **Prefix** $F_{\ell-1} : \mathcal{X} \to \mathbb{R}^{T \times d}$ maps the tokens to the hidden state entering layer ℓ ($F_0 = E_{\text{mm}}$).
- **Hidden state** (residual stream) after layer ℓ: $h^{(\ell)} = F_\ell(x) = \mathcal{B}^{(\ell)}(h^{(\ell-1)}) \in \mathbb{R}^{T \times d}$, with $h^{(0)} = E_{\text{mm}}(x)$.
- **Inside layer ℓ:** the attention write $a^{(\ell)} = \text{Mixer}_\ell\big(\text{RMSNorm}_1(h^{(\ell-1)})\big)$, the mid-block state $\hat{h}^{(\ell)} = h^{(\ell-1)} + a^{(\ell)}$, the feed-forward input $u^{(\ell)} = \text{RMSNorm}_2(\hat{h}^{(\ell)})$, the pre-activations $\hat{x}^{(\ell)} = u^{(\ell)} W_{\text{gate}}^{(\ell)\top} \in \mathbb{R}^{T \times d_{ff}}$ and the neuron outputs $\eta^{(\ell)}(u^{(\ell)}) \in \mathbb{R}^{T \times d_{ff}}$, and the feed-forward write $m^{(\ell)} = \text{FFN}_\ell(u^{(\ell)})$.
- **Suffix** $G_{\ell+1} : \mathbb{R}^{T \times d} \to \mathbb{R}^{T \times V}$ maps $h^{(\ell)}$ to the logits ($G_{L+1} = \Psi$).

Residual stream. Because every sub-layer only *adds* to the hidden state, unrolling the composition gives the final hidden state as the embedding plus everything the layers wrote:

$$h^{(L)} = h^{(0)} + \sum_{\ell=1}^{L} \big(a^{(\ell)} + m^{(\ell)}\big) \tag{16}$$

So $f_\theta(x) = \Psi(h^{(L)})$: each layer reads the current stream and adds a correction to it.

2.7 Local linear representation

A ReLU network is piecewise affine: inside a polytope cell it collapses to one matrix. This model is not, because softmax, SiLU and the normalisations are smooth. Its local behaviour is described by the first-order (Jacobian) linearisation instead. Tokens are discrete, so we perturb the embedded input $h^{(0)}$.

Definition 2.14 (Local linear response)**.** Let $J_\ell = D\mathcal{B}^{(\ell)}(h^{(\ell-1)})$ be the Jacobian of block ℓ at the current hidden state (tensors flattened to vectors). The chain rule applied to the residual sub-layers $(\text{id} + g_{\ell,i})$ gives

$$J_\ell = (I + Dg_{\ell,2})(I + Dg_{\ell,1}), \tag{17}$$

where $g_{\ell,1}$ is the attention branch, $g_{\ell,2}$ the feed-forward branch and each Dg is taken at that branch's input. For a small perturbation δ of $h^{(0)}$, the response of the network decomposed around layer ℓ is

$$f_\theta(h^{(0)} + \delta) - f_\theta(h^{(0)}) \approx G_x^{(\ell+1)} \cdot J_\ell \cdot F_x^{(\ell-1)} \cdot \delta, \tag{18}$$

with $F_x^{(\ell-1)} = J_{\ell-1} \cdots J_1$ (prefix) and $G_x^{(\ell+1)} = J_\Psi\, J_L \cdots J_{\ell+1}$ (suffix). Multiplying out the products of $(I + Dg)$ terms writes $J_\Psi J_L \cdots J_1$ as a sum over *paths*: the identity path straight from embedding to head, plus one term for every subset of sub-layers a signal passes through.

2.8 Summary of notation

Symbol	Meaning
$f_\theta : \mathcal{X} \to \mathcal{Y}$	the whole model, $\{1..V\}^T \to \mathbb{R}^{T \times V}$
$E_{\text{mm}},\ \Psi$	embedding and output head
$\text{Enc}_\text{V},\ \text{Enc}_\text{A},\ \text{Proj}$	vision / audio encoders and their projectors into the text width
$\mathcal{B}^{(\ell)},\ \theta_\ell$	block ℓ and its parameters, $\ell = 1, \dots, L$
$\mathcal{B}_c,\ c(\ell)$	block type c; type of layer ℓ
$\mathcal{B}^{\circ n}$	n consecutive blocks of one type (each with its own weights)
$(\text{id} + g)$	residual sub-layer $h \mapsto h + g(h)$
$h^{(\ell)}$	hidden state (residual stream) after layer ℓ; $h^{(0)}$ = embeddings
$a^{(\ell)},\ m^{(\ell)}$	what the attention / feed-forward sub-layer of layer ℓ adds to the stream
$\eta_j^{(\ell)}$	j-th feed-forward neuron of layer ℓ
$\mathcal{V}(\eta; x, t),\ \mathcal{S}(\eta; x, t)$	pre-activation value and active/inactive state of a neuron

continued on next page

Symbol	Meaning
$P_h^{(\ell)}$	attention pattern of head h in layer ℓ ($T \times T$)
$F_{\ell-1},\ G_{\ell+1}$	prefix (embedding to layer $\ell-1$) and suffix (layer $\ell+1$ to logits)
$J_\ell,\ F_x^{(\ell-1)},\ G_x^{(\ell+1)}$	Jacobian of block ℓ, and of the prefix / suffix at input x
$p(x),\ z(x)$	next-token distribution and greedy next-token prediction

3 Forward Pass in Pseudocode

Algorithm 1 Qwen3.5-0.8B: full forward pass

Require: token ids x of shape $B \times T$; optional images / audio

$h \leftarrow E(x)$ ▷ $h^{(0)}$: one vector per token, $B{\times}T{\times}1024$
if image input present **then**
  $e \leftarrow \mathrm{Enc_V}(\text{image})$ ▷ vision encoder: 12 layers, Algorithm 2
  $h[\text{image token slots}] \leftarrow \mathrm{Proj}(e)$ ▷ project into text width 1024
end if
$\ell \leftarrow 0$
for $r \leftarrow 1$ **to** 6 **do** ▷ layers 1–24: pattern of 4 blocks, repeated 6 times
  for $j \leftarrow 1$ **to** 3 **do**
    $\ell \leftarrow \ell + 1;\ h \leftarrow \mathcal{B}_\mathrm{L}(h;\, \theta_\ell)$ ▷ Linear-attention (Gated DeltaNet) block
  end for
  $\ell \leftarrow \ell + 1;\ h \leftarrow \mathcal{B}_\mathrm{F}(h;\, \theta_\ell)$ ▷ Full (global) attention block
end for
$h \leftarrow \mathrm{RMSNorm}_{\mathrm{final}}(h)$ ▷ $h = h^{(L)}$; this and the next step are Ψ
$y \leftarrow h\, W_E^\top$ ▷ logits $B{\times}T{\times}248,320$, embedding matrix re-used
return y ▷ $\mathrm{softmax}(y)$ = next-token probabilities

Algorithm 2 Vision encoder

$h \leftarrow \mathrm{PatchEmbed}(\text{pixels}) + \text{position embedding}$ ▷ one vector per image patch
for $\ell \leftarrow 1$ **to** 12 **do** ▷ 12 blocks of type $\mathcal{B}_\mathrm{V}$
  $h \leftarrow \mathcal{B}_\mathrm{V}(h;\, \theta_\ell)$ ▷ Vision encoder block; result is $h^{(\ell)}$
end for
$h \leftarrow \mathrm{Post}(h)$ ▷ final norm / pooling / merging, see Table 3
return h

Algorithm 3 Block $\mathcal{B}_\mathrm{L}(h)$ — Linear-attention (Gated DeltaNet) block

Require: h of shape $B \times T \times 1024$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: Gated DeltaNet (linear attention, fixed-size memory)
$q, k, v \leftarrow \mathrm{split}(\mathrm{SiLU}(\mathrm{CausalConv1d}(uW_{qkv}^\top)))$ ▷ 16 q/k heads of 128, 16 v heads of 128
$z \leftarrow uW_z^\top;\ \beta \leftarrow \sigma(uW_b^\top);\ g \leftarrow -e^{A_{\log}}\,\mathrm{softplus}(uW_a^\top + b_{dt})$ ▷ gate, write strength, decay
for $t \leftarrow 1$ **to** T **do** ▷ conceptually sequential; computed in chunks
  $S \leftarrow e^{g_t} S\,(I - \beta_t k_t k_t^\top) + \beta_t v_t k_t^\top$ ▷ forget a bit, overwrite what k_t points at
  $o_t \leftarrow S\, q_t$ ▷ read
end for
$m \leftarrow \mathrm{RMSNorm}(o) \odot \mathrm{SiLU}(z)\ W_{\mathrm{out}}^\top$ ▷ back to $B{\times}T{\times}1024$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{SiLU}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big) W_{\mathrm{down}}^\top$ ▷ expand to 3584, gate, shrink back to 1024
$h \leftarrow r + m$ ▷ add residual
return h

Algorithm 4 Block $\mathcal{B}_{\mathrm{F}}(h)$ — Full (global) attention block

Require: h of shape $B \times T \times 1024$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{RMSNorm}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q \leftarrow uW_Q^\top$ ▷ $B{\times}T{\times}8{\times}256$ (8 query heads); second half is a gate G
$K \leftarrow uW_K^\top;\ V \leftarrow uW_V^\top$ ▷ $B{\times}T{\times}2{\times}256$ (2 key/value heads, each shared by 4 query heads)
$Q \leftarrow \mathrm{RMSNorm}(Q);\ K \leftarrow \mathrm{RMSNorm}(K)$ ▷ normalise each head vector (stabilises training)
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ encode position by rotating vector pairs
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: causal: a token sees only earlier tokens
$O \leftarrow PV$ ▷ $B{\times}T{\times}8{\times}256$
$O \leftarrow O \odot \sigma(G)$ ▷ output gate decides how much of each head to keep
$m \leftarrow O\,W_O^\top$ ▷ concatenate heads, project back to $B{\times}T{\times}1024$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{RMSNorm}(h)$
$m \leftarrow \big(\mathrm{SiLU}(uW_{\mathrm{gate}}^\top) \odot uW_{\mathrm{up}}^\top\big)W_{\mathrm{down}}^\top$ ▷ expand to 3584, gate, shrink back to 1024
$h \leftarrow r + m$ ▷ add residual
return h

Algorithm 5 Block $\mathcal{B}_{\mathrm{V}}(h)$ — Vision encoder block

Require: h of shape $B \times T \times 768$

$r \leftarrow h$ ▷ keep the input for the residual (skip) connection
$u \leftarrow \mathrm{LN}(h)$ ▷ normalise every token vector
token mixing: multi-head self-attention
$Q, K, V \leftarrow \mathrm{split}(uW_{qkv}^\top + b)$ ▷ each $B{\times}T{\times}12{\times}64$
$Q, K \leftarrow \mathrm{RoPE}(Q, K)$ ▷ 2-D rotary position (row, column)
$P \leftarrow \mathrm{softmax}(s\,QK^\top + M)$ ▷ mask: every patch/frame sees every other
$O \leftarrow PV$ ▷ $B{\times}T{\times}12{\times}64$
$m \leftarrow O\,W_O^\top + b_O$ ▷ concatenate heads, project back to $B{\times}T{\times}768$
$h \leftarrow r + m$ ▷ add residual
channel mixing: feed-forward network (each token independently)
$r \leftarrow h;\ u \leftarrow \mathrm{LN}(h)$
$m \leftarrow \mathrm{GELU}_{\tanh}(uW_1^\top + b_1)W_2^\top + b_2$ ▷ expand to 3072, shrink back to 768
$h \leftarrow r + m$ ▷ add residual
return h

4 Compact Architecture (repeated layers written once)

4.1 Modules outside the repeated stacks

Table 3: Weights that appear once

Weight	Type	Shape	Params
`model.visual.patch_embed.proj.weight`	Conv3d	$768 \times 3 \times 2 \times 16 \times 16$	1,179,648
`model.visual.patch_embed.proj.bias`	Conv3d	768	768
`model.visual.pos_embed.weight`	Embedding	2304×768	1,769,472
`model.visual.merger.norm.weight`	LayerNorm	768	768
`model.visual.merger.norm.bias`	LayerNorm	768	768
`model.visual.merger.linear_fc1.weight`	Linear	3072×3072	9,437,184
`model.visual.merger.linear_fc1.bias`	Linear	3072	3,072
`model.visual.merger.linear_fc2.weight`	Linear	1024×3072	3,145,728
`model.visual.merger.linear_fc2.bias`	Linear	1024	1,024
`model.language_model.embed_tokens.weight`	Embedding	248320×1024	254,279,680
`model.language_model.norm.weight`	Qwen3_5RMSNorm	1024	1,024

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Weight	Type	Shape	Params
`lm_head.weight` (tied to `model.language_model.embed_tokens.weight`)	Linear	248320×1024	0 (shared)

4.2 Language-model decoder: `model.language_model.layers`

Table 4: $\mathcal{B}_{\mathrm{L}}$: Linear-attention (Gated DeltaNet) block — layers 0–2, 4–6, 8–10, 12–14, 16–18, 20–22; weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`linear_attn.dt_bias`	Qwen3_5GatedDeltaNet	16	16
`linear_attn.A_log`	Qwen3_5GatedDeltaNet	16	16
`linear_attn.conv1d.weight`	Conv1d	$6144 \times 1 \times 4$	24,576
`linear_attn.norm.weight`	Qwen3_5RMSNormGated	128	128
`linear_attn.out_proj.weight`	Linear	1024×2048	2,097,152
`linear_attn.in_proj_qkv.weight`	Linear	6144×1024	6,291,456
`linear_attn.in_proj_z.weight`	Linear	2048×1024	2,097,152
`linear_attn.in_proj_b.weight`	Linear	16×1024	16,384
`linear_attn.in_proj_a.weight`	Linear	16×1024	16,384
`mlp.gate_proj.weight`	Linear	3584×1024	3,670,016
`mlp.up_proj.weight`	Linear	3584×1024	3,670,016
`mlp.down_proj.weight`	Linear	1024×3584	3,670,016
`input_layernorm.weight`	Qwen3_5RMSNorm	1024	1,024
`post_attention_layernorm.weight`	Qwen3_5RMSNorm	1024	1,024
Per block			**21,555,360**
× 18 blocks			**387,996,480**

Table 5: $\mathcal{B}_{\mathrm{F}}$: Full (global) attention block — layers 3, 7, …, 23 (every 4th); weight names are relative to `model.language_model.layers.`ℓ

Weight (relative)	Type	Shape	Params
`self_attn.q_proj.weight`	Linear	4096×1024	4,194,304
`self_attn.k_proj.weight`	Linear	512×1024	524,288
`self_attn.v_proj.weight`	Linear	512×1024	524,288
`self_attn.o_proj.weight`	Linear	1024×2048	2,097,152
`self_attn.q_norm.weight`	Qwen3_5RMSNorm	256	256
`self_attn.k_norm.weight`	Qwen3_5RMSNorm	256	256
`mlp.gate_proj.weight`	Linear	3584×1024	3,670,016
`mlp.up_proj.weight`	Linear	3584×1024	3,670,016
`mlp.down_proj.weight`	Linear	1024×3584	3,670,016
`input_layernorm.weight`	Qwen3_5RMSNorm	1024	1,024
`post_attention_layernorm.weight`	Qwen3_5RMSNorm	1024	1,024
Per block			**18,352,640**
× 6 blocks			**110,115,840**

4.3 Vision encoder: `model.visual.blocks`

Table 6: $\mathcal{B}_V$: Vision encoder block — layers 0–11; weight names are relative to `model.visual.blocks.`ℓ

Weight (relative)	Type	Shape	Params
`norm1.weight`	LayerNorm	768	768
`norm1.bias`	LayerNorm	768	768
`norm2.weight`	LayerNorm	768	768
`norm2.bias`	LayerNorm	768	768
`attn.qkv.weight`	Linear	2304×768	1,769,472
`attn.qkv.bias`	Linear	2304	2,304
`attn.proj.weight`	Linear	768×768	589,824
`attn.proj.bias`	Linear	768	768
`mlp.linear_fc1.weight`	Linear	3072×768	2,359,296
`mlp.linear_fc1.bias`	Linear	3072	3,072
`mlp.linear_fc2.weight`	Linear	768×3072	2,359,296
`mlp.linear_fc2.bias`	Linear	768	768
Per block			**7,087,872**
× 12 blocks			**85,054,464**

4.4 Checkpoint-only modules

Multi-token-prediction (MTP) head (`mtp.*`). An extra one-layer transformer that predicts the token after next. It concatenates RMSNorm(embedding of token $t{+}1$) with RMSNorm(final hidden state at t), projects $2d \to d$ (`fc`), runs one full-attention decoder layer, then a norm and the shared LM head. Serving engines (vLLM/SGLang) use it for speculative decoding. The transformers class does not instantiate it, so it is not counted in the total above.

Tensor	Shape	Dtype	Params
`mtp.fc.weight`	1024×2048	bf16	2,097,152
`mtp.layers.0.input_layernorm.weight`	1024	bf16	1,024
`mtp.layers.0.mlp.down_proj.weight`	1024×3584	bf16	3,670,016
`mtp.layers.0.mlp.gate_proj.weight`	3584×1024	bf16	3,670,016
`mtp.layers.0.mlp.up_proj.weight`	3584×1024	bf16	3,670,016
`mtp.layers.0.post_attention_layernorm.weight`	1024	bf16	1,024
`mtp.layers.0.self_attn.k_norm.weight`	256	bf16	256
`mtp.layers.0.self_attn.k_proj.weight`	512×1024	bf16	524,288
`mtp.layers.0.self_attn.o_proj.weight`	1024×2048	bf16	2,097,152
`mtp.layers.0.self_attn.q_norm.weight`	256	bf16	256
`mtp.layers.0.self_attn.q_proj.weight`	4096×1024	bf16	4,194,304
`mtp.layers.0.self_attn.v_proj.weight`	512×1024	bf16	524,288
`mtp.norm.weight`	1024	bf16	1,024
`mtp.pre_fc_norm_embedding.weight`	1024	bf16	1,024
`mtp.pre_fc_norm_hidden.weight`	1024	bf16	1,024

5 What Each Layer Does

One row per *kind* of layer present in this model. "Appears as" gives the module names (layer indices removed).

Layer (appears as)	What it does
LANGUAGE MODEL	
Token embedding `embed_tokens`	Lookup table with one learned vector per vocabulary token. Turns token ids into the vectors the network computes with (Gemma multiplies them by $\sqrt{d}$).
Query projection (q_proj) `self_attn.q_proj`	Turns each token vector into query vectors, one per head: "what am I looking for?". Here it also outputs an equally sized gate for the attention output (hence the doubled width).
Query norm (q_norm) `self_attn.q_norm`	RMSNorm applied to every query head vector before RoPE. Keeps attention logits in a sane range (QK-norm).

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Layer (appears as)	What it does
Key projection (k_proj) `self_attn.k_proj`	Turns each token into key vectors: "what do I contain?". Queries are compared with keys to decide where to attend. With grouped-query attention there are fewer key heads than query heads.
Key norm (k_norm) `self_attn.k_norm`	RMSNorm applied to every key head vector before RoPE.
Value projection (v_proj) `self_attn.v_proj`	Turns each token into value vectors: the information actually copied to tokens that attend to it.
DeltaNet output projection `linear_attn.out_proj`	Mixes the heads back into one hidden-size vector.
Output projection (o_proj) `self_attn.o_proj`	Concatenates all heads' results and mixes them back into one hidden-size vector per token.
DeltaNet Q/K/V projection `linear_attn.in_proj_qkv`	Produces queries, keys and values for the linear-attention layer in one matrix.
DeltaNet output-gate projection `linear_attn.in_proj_z`	Produces gate z. The normalised output is multiplied by SiLU(z).
DeltaNet write-strength projection `linear_attn.in_proj_b`	Produces $\beta = \sigma(\cdot)$ per head: how strongly the current token overwrites the memory.
DeltaNet decay projection `linear_attn.in_proj_a`	Produces the input-dependent forgetting rate (with A_log and dt_bias).
Short causal convolution `linear_attn.conv1d`	Depthwise 1-D convolution over the last 4 tokens of Q/K/V. Gives cheap local context before the recurrent memory update.
A_log (decay base) `layers.linear_attn`	Per-head learned log decay rate. $-e^{A_{\log}}$ sets how quickly old memory fades.
dt_bias (decay bias) `layers.linear_attn`	Per-head bias added before the softplus that controls the decay step.
Gated output RMSNorm `linear_attn.norm`	Normalises each head's memory read-out and multiplies it by SiLU(z).
Gate projection (gate_proj) `mlp.gate_proj`	Expands the token vector to the MLP width and passes it through SiLU/GELU. The result acts as a soft on/off switch for each hidden unit.
Up projection (up_proj) `mlp.up_proj`	Expands the token vector to the MLP width (the "content" path). It is multiplied element-wise by the gate (SwiGLU/GeGLU).
Down projection (down_proj) `mlp.down_proj`	Shrinks the gated hidden vector back to the model width so it can be added to the residual stream.
Input norm (input_layernorm) `layers.input_layernorm`	RMSNorm applied to the residual stream before attention. Rescales each token vector to unit RMS, then multiplies by a learned per-channel weight.
post_attention_layernorm `layers.post_attention_layernorm`	Despite the name, this is the pre-MLP norm: RMSNorm on the residual stream after attention, feeding the MLP.
Final norm `norm`	RMSNorm applied once after the last decoder layer, just before the LM head.
VISION ENCODER	
3-D patch embedding (Conv3d) `patch_embed.proj`	Cuts images/videos into 2-frame × 16 × 16-pixel cubes and maps each cube to one vector.
Position embedding `visual.pos_embed`	Learned position vectors, interpolated to the image's patch grid.
Pre-attention LayerNorm `blocks.norm1`	LayerNorm before vision self-attention.
Pre-MLP LayerNorm `blocks.norm2`	LayerNorm before the vision MLP.
Fused Q/K/V projection `attn.qkv`	One matrix that produces queries, keys and values at once (split afterwards).
Attention output projection `attn.proj`	Mixes the concatenated head outputs back to the model width.

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Layer (appears as)	What it does
MLP up-projection (fc1) `mlp.linear_fc1`	Expands each vector to the wider MLP size before the GELU non-linearity.
MLP down-projection (fc2) `mlp.linear_fc2`	Shrinks back to the model width.
Merger LayerNorm `merger.norm`	Normalises patch features before 2×2 neighbouring patches are merged.
Merger MLP layer 1 `merger.linear_fc1`	Takes 4 neighbouring patches concatenated (2×2 merge, so 4× fewer image tokens) and mixes them.
Merger MLP layer 2 (to text width) `merger.linear_fc2`	Projects merged patches into the language model's hidden size.

6 Parameter Count

Table 9: Parameters by component

Component	Parameters	% of total
MLP / MoE	264,241,152	31.0%
Token embedding	254,279,680	29.8%
Attention / token mixer	233,822,016	27.4%
Vision encoder	100,592,896	11.8%
Norms	50,176	0.0%
Total	**852,985,920**	**100.0%**

Table 10: Parameters by block type

Block type	Count	Params / block	Total
$\mathcal{B}_{\mathrm{L}}$ Linear-attention (Gated DeltaNet) block	18	21,555,360	387,996,480
$\mathcal{B}_{\mathrm{F}}$ Full (global) attention block	6	18,352,640	110,115,840
$\mathcal{B}_{\mathrm{V}}$ Vision encoder block	12	7,087,872	85,054,464
Modules outside stacks	1	–	269,819,136
Total			**852,985,920**

Appendix A Full Layer-by-Layer Table

Every weight tensor with every repetition written out (the original table).

Layer path	Type	Shape	Dtype	Params
`lm_head.weight` *(tied)*	Linear	248320 × 1024	bfloat16	254,279,680
`model`	Qwen3_5Model	–	–	–
`model.language_model`	Qwen3_5TextModel	–	–	–
`model.language_model.embed_tokens.weight`	Embedding	248320 × 1024	bfloat16	254,279,680
`model.language_model.layers.0`	Qwen3_5DecoderLayer	–	–	–
`model.language_model.layers.0.input_layernorm.weight`	Qwen3_5RMSNorm	1024	bfloat16	1,024
`model.language_model.layers.0.linear_attn.A_log`	Qwen3_5GatedDeltaNet	16	bfloat16	16
`model.language_model.layers.0.linear_attn`	Qwen3_5GatedDeltaNet	–	–	–
`model.language_model.layers.0.linear_attn.conv1d.weight`	Conv1d	6144 × 1 × 4	bfloat16	24,576
`model.language_model.layers.0.linear_attn.dt_bias`	Qwen3_5GatedDeltaNet	16	bfloat16	16
`model.language_model.layers.0.linear_attn.in_proj_a.weight`	Linear	16 × 1024	bfloat16	16,384
`model.language_model.layers.0.linear_attn.in_proj_b.weight`	Linear	16 × 1024	bfloat16	16,384
`model.language_model.layers.0.linear_attn.in_proj_qkv.weight`	Linear	6144 × 1024	bfloat16	6,291,456

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.0.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.0.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.0.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.0.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.0.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.0.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.0.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.0.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.1	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.1.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.1.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.1.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.1.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.1.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.1.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.1.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.1.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.1.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.1.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.1.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.1.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.1.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.1.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.1.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.1.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.2	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.2.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.2.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.2.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.2.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.2.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.2.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.2.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.2.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.2.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.2.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.2.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.2.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.2.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.2.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.2.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.2.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.3	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.3.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.3.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.3.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.3.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.3.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.3.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.3.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.3.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.3.self_attn.k_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.3.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.3.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.3.self_attn.q_proj.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.language_model.layers.3.self_attn.v_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.4	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.4.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.4.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.4.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.4.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.4.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.4.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.4.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.4.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.4.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.4.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.4.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.4.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.4.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.4.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.4.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.4.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.5	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.5.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.5.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.5.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.5.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.5.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.5.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.5.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.5.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.5.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.5.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.5.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.5.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.5.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.5.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.5.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.5.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.6	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.6.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.6.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.6.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.6.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.6.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.6.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.6.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.6.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.6.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.6.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.6.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.6.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.6.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.6.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.6.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.6.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.7	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.7.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.7.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.7.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.7.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.7.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.7.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.7.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.7.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.7.self_attn.k_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.7.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.7.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.7.self_attn.q_proj.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.language_model.layers.7.self_attn.v_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.8	Qwen3_5DecoderLayer	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.8.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.8.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.8.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.8.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.8.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.8.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.8.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.8.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.8.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.8.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.8.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.8.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.8.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.8.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.8.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.8.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.9	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.9.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.9.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.9.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.9.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.9.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.9.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.9.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.9.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.9.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.9.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.9.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.9.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.9.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.9.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.9.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.9.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.10	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.10.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.10.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.10.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.10.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.10.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.10.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.10.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.10.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.10.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.10.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.10.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.10.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.10.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.10.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.10.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.10.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.11	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.11.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.11.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.11.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.11.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.11.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.11.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.11.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.11.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.11.self_attn.k_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.11.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.11.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.11.self_attn.q_proj.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.language_model.layers.11.self_attn.v_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.12	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.12.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.12.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.12.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.12.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.12.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.12.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.12.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.12.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.12.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.12.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.12.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.12.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.12.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.12.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.12.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.12.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.13	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.13.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.13.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.13.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.13.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.13.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.13.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.13.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.13.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.13.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.13.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.13.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.13.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.13.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.13.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.13.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.13.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.14	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.14.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.14.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.14.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.14.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.14.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.14.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.14.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.14.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.14.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.14.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.14.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.14.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.14.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.14.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.14.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.14.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.15	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.15.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.15.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.15.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.15.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.15.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.15.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.15.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.15.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.15.self_attn.k_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.15.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.15.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.15.self_attn.q_proj.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.language_model.layers.15.self_attn.v_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.16	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.16.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.16.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.16.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.16.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.16.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.16.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.16.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.16.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.16.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.16.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.16.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.16.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.16.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.16.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.16.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.16.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.17	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.17.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.17.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.17.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.17.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.17.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.17.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.17.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.17.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.17.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.17.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.17.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.17.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.17.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.17.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.17.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.17.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.18	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.18.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.18.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.18.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.18.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.18.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.18.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.18.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.18.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.18.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.18.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.18.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.18.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.18.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.18.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.18.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.18.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.19	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.19.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.19.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.19.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.19.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.19.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.19.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.19.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.19.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.19.self_attn.k_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.19.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.19.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.19.self_attn.q_proj.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.language_model.layers.19.self_attn.v_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.20	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.20.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.20.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.20.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.20.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.20.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.20.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.20.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.20.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.20.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.20.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.20.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.20.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.20.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.20.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.20.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.20.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.21	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.21.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.21.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.21.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.21.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.21.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.21.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.21.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.21.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.21.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.21.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.21.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.21.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.21.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.21.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.21.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.21.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.22	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.22.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.22.linear_attn.A_log	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.22.linear_attn	Qwen3_5GatedDeltaNet	–	–	–
model.language_model.layers.22.linear_attn.conv1d.weight	Conv1d	6144 × 1 × 4	bfloat16	24,576
model.language_model.layers.22.linear_attn.dt_bias	Qwen3_5GatedDeltaNet	16	bfloat16	16
model.language_model.layers.22.linear_attn.in_proj_a.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.22.linear_attn.in_proj_b.weight	Linear	16 × 1024	bfloat16	16,384
model.language_model.layers.22.linear_attn.in_proj_qkv.weight	Linear	6144 × 1024	bfloat16	6,291,456
model.language_model.layers.22.linear_attn.in_proj_z.weight	Linear	2048 × 1024	bfloat16	2,097,152
model.language_model.layers.22.linear_attn.norm.weight	Qwen3_5RMSNormGated	128	bfloat16	128
model.language_model.layers.22.linear_attn.out_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.22.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.22.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016

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Layer path	Type	Shape	Dtype	Params
model.language_model.layers.22.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.22.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.22.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.23	Qwen3_5DecoderLayer	–	–	–
model.language_model.layers.23.input_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.23.mlp	Qwen3_5MLP	–	–	–
model.language_model.layers.23.mlp.down_proj.weight	Linear	1024 × 3584	bfloat16	3,670,016
model.language_model.layers.23.mlp.gate_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.23.mlp.up_proj.weight	Linear	3584 × 1024	bfloat16	3,670,016
model.language_model.layers.23.post_attention_layernorm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.language_model.layers.23.self_attn	Qwen3_5Attention	–	–	–
model.language_model.layers.23.self_attn.k_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.23.self_attn.k_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.layers.23.self_attn.o_proj.weight	Linear	1024 × 2048	bfloat16	2,097,152
model.language_model.layers.23.self_attn.q_norm.weight	Qwen3_5RMSNorm	256	bfloat16	256
model.language_model.layers.23.self_attn.q_proj.weight	Linear	4096 × 1024	bfloat16	4,194,304
model.language_model.layers.23.self_attn.v_proj.weight	Linear	512 × 1024	bfloat16	524,288
model.language_model.norm.weight	Qwen3_5RMSNorm	1024	bfloat16	1,024
model.visual	Qwen3_5VisionModel	–	–	–
model.visual.blocks.0	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.0.attn	Qwen3_5VisionAttention	–	–	–
model.visual.blocks.0.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.0.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.0.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.0.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.0.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.0.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.0.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.0.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.0.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.0.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.0.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.0.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.0.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.1	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.1.attn	Qwen3_5VisionAttention	–	–	–
model.visual.blocks.1.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.1.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.1.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.1.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.1.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.1.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.1.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.1.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.1.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.1.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.1.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.1.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.1.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.2	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.2.attn	Qwen3_5VisionAttention	–	–	–
model.visual.blocks.2.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.2.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.2.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.2.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.2.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.2.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072

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Layer path	Type	Shape	Dtype	Params
model.visual.blocks.2.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.2.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.2.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.2.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.2.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.2.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.2.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.3	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.3.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.3.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.3.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.3.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.3.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.3.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.3.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.3.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.3.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.3.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.3.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.3.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.3.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.3.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.4	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.4.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.4.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.4.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.4.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.4.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.4.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.4.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.4.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.4.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.4.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.4.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.4.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.4.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.4.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.5	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.5.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.5.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.5.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.5.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.5.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.5.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.5.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.5.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.5.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.5.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.5.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.5.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.5.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.5.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.6	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.6.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.6.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.6.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.6.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.6.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.6.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.6.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.6.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.6.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.6.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.6.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.6.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.6.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.6.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.7	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.7.attn	Qwen3_5Vision Attention	–	–	–

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Layer path	Type	Shape	Dtype	Params
model.visual.blocks.7.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.7.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.7.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.7.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.7.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.7.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.7.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.7.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.7.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.7.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.7.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.7.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.7.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.8	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.8.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.8.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.8.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.8.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.8.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.8.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.8.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.8.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.8.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.8.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.8.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.8.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.8.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.8.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.9	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.9.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.9.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.9.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.9.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.9.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.9.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.9.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.9.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.9.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.9.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.9.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.9.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.9.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.9.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.10	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.10.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.10.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.10.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.10.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.10.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.10.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.10.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.10.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.10.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.10.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.10.norm1.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.10.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.10.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.10.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.11	Qwen3_5VisionBlock	–	–	–
model.visual.blocks.11.attn	Qwen3_5Vision Attention	–	–	–
model.visual.blocks.11.attn.proj.bias	Linear	768	bfloat16	768
model.visual.blocks.11.attn.proj.weight	Linear	768 × 768	bfloat16	589,824
model.visual.blocks.11.attn.qkv.bias	Linear	2304	bfloat16	2,304
model.visual.blocks.11.attn.qkv.weight	Linear	2304 × 768	bfloat16	1,769,472
model.visual.blocks.11.mlp	Qwen3_5VisionMLP	–	–	–
model.visual.blocks.11.mlp.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.blocks.11.mlp.linear_fc1.weight	Linear	3072 × 768	bfloat16	2,359,296
model.visual.blocks.11.mlp.linear_fc2.bias	Linear	768	bfloat16	768
model.visual.blocks.11.mlp.linear_fc2.weight	Linear	768 × 3072	bfloat16	2,359,296
model.visual.blocks.11.norm1.bias	LayerNorm	768	bfloat16	768

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Layer path	Type	Shape	Dtype	Params
model.visual.blocks.11.norm1.weight	LayerNorm	768	bfloat16	768
model.visual.blocks.11.norm2.bias	LayerNorm	768	bfloat16	768
model.visual.blocks.11.norm2.weight	LayerNorm	768	bfloat16	768
model.visual.merger	Qwen3_5VisionPatch Merger	–	–	–
model.visual.merger.linear_fc1.bias	Linear	3072	bfloat16	3,072
model.visual.merger.linear_fc1.weight	Linear	3072 × 3072	bfloat16	9,437,184
model.visual.merger.linear_fc2.bias	Linear	1024	bfloat16	1,024
model.visual.merger.linear_fc2.weight	Linear	1024 × 3072	bfloat16	3,145,728
model.visual.merger.norm.bias	LayerNorm	768	bfloat16	768
model.visual.merger.norm.weight	LayerNorm	768	bfloat16	768
model.visual.patch_embed	Qwen3_5VisionPatch Embed	–	–	–
model.visual.patch_embed.proj.bias	Conv3d	768	bfloat16	768
model.visual.patch_embed.proj.weight	Conv3d	768 × 3 × 2 × 16 × 16	bfloat16	1,179,648
model.visual.pos_embed.weight	Embedding	2304 × 768	bfloat16	1,769,472

Appendix B Per-Layer Parameter Count

Module	Params	Running total
model.language_model.embed_tokens	254,279,680	254,279,680
model.language_model.layers.0.input_layernorm	1,024	254,280,704
model.language_model.layers.0.linear_attn	32	254,280,736
model.language_model.layers.0.linear_attn.conv1d	24,576	254,305,312
model.language_model.layers.0.linear_attn.in_proj_a	16,384	254,321,696
model.language_model.layers.0.linear_attn.in_proj_b	16,384	254,338,080
model.language_model.layers.0.linear_attn. in_proj_qkv	6,291,456	260,629,536
model.language_model.layers.0.linear_attn.in_proj_z	2,097,152	262,726,688
model.language_model.layers.0.linear_attn.norm	128	262,726,816
model.language_model.layers.0.linear_attn.out_proj	2,097,152	264,823,968
model.language_model.layers.0.mlp.down_proj	3,670,016	268,493,984
model.language_model.layers.0.mlp.gate_proj	3,670,016	272,164,000
model.language_model.layers.0.mlp.up_proj	3,670,016	275,834,016
model.language_model.layers.0. post_attention_layernorm	1,024	275,835,040
model.language_model.layers.1.input_layernorm	1,024	275,836,064
model.language_model.layers.1.linear_attn	32	275,836,096
model.language_model.layers.1.linear_attn.conv1d	24,576	275,860,672
model.language_model.layers.1.linear_attn.in_proj_a	16,384	275,877,056
model.language_model.layers.1.linear_attn.in_proj_b	16,384	275,893,440
model.language_model.layers.1.linear_attn. in_proj_qkv	6,291,456	282,184,896
model.language_model.layers.1.linear_attn.in_proj_z	2,097,152	284,282,048
model.language_model.layers.1.linear_attn.norm	128	284,282,176
model.language_model.layers.1.linear_attn.out_proj	2,097,152	286,379,328
model.language_model.layers.1.mlp.down_proj	3,670,016	290,049,344
model.language_model.layers.1.mlp.gate_proj	3,670,016	293,719,360
model.language_model.layers.1.mlp.up_proj	3,670,016	297,389,376
model.language_model.layers.1. post_attention_layernorm	1,024	297,390,400
model.language_model.layers.2.input_layernorm	1,024	297,391,424
model.language_model.layers.2.linear_attn	32	297,391,456
model.language_model.layers.2.linear_attn.conv1d	24,576	297,416,032
model.language_model.layers.2.linear_attn.in_proj_a	16,384	297,432,416
model.language_model.layers.2.linear_attn.in_proj_b	16,384	297,448,800
model.language_model.layers.2.linear_attn. in_proj_qkv	6,291,456	303,740,256
model.language_model.layers.2.linear_attn.in_proj_z	2,097,152	305,837,408
model.language_model.layers.2.linear_attn.norm	128	305,837,536

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Module	Params	Running total
model.language_model.layers.2.linear_attn.out_proj	2,097,152	307,934,688
model.language_model.layers.2.mlp.down_proj	3,670,016	311,604,704
model.language_model.layers.2.mlp.gate_proj	3,670,016	315,274,720
model.language_model.layers.2.mlp.up_proj	3,670,016	318,944,736
model.language_model.layers.2. post_attention_layernorm	1,024	318,945,760
model.language_model.layers.3.input_layernorm	1,024	318,946,784
model.language_model.layers.3.mlp.down_proj	3,670,016	322,616,800
model.language_model.layers.3.mlp.gate_proj	3,670,016	326,286,816
model.language_model.layers.3.mlp.up_proj	3,670,016	329,956,832
model.language_model.layers.3. post_attention_layernorm	1,024	329,957,856
model.language_model.layers.3.self_attn.k_norm	256	329,958,112
model.language_model.layers.3.self_attn.k_proj	524,288	330,482,400
model.language_model.layers.3.self_attn.o_proj	2,097,152	332,579,552
model.language_model.layers.3.self_attn.q_norm	256	332,579,808
model.language_model.layers.3.self_attn.q_proj	4,194,304	336,774,112
model.language_model.layers.3.self_attn.v_proj	524,288	337,298,400
model.language_model.layers.4.input_layernorm	1,024	337,299,424
model.language_model.layers.4.linear_attn	32	337,299,456
model.language_model.layers.4.linear_attn.conv1d	24,576	337,324,032
model.language_model.layers.4.linear_attn.in_proj_a	16,384	337,340,416
model.language_model.layers.4.linear_attn.in_proj_b	16,384	337,356,800
model.language_model.layers.4.linear_attn. in_proj_qkv	6,291,456	343,648,256
model.language_model.layers.4.linear_attn.in_proj_z	2,097,152	345,745,408
model.language_model.layers.4.linear_attn.norm	128	345,745,536
model.language_model.layers.4.linear_attn.out_proj	2,097,152	347,842,688
model.language_model.layers.4.mlp.down_proj	3,670,016	351,512,704
model.language_model.layers.4.mlp.gate_proj	3,670,016	355,182,720
model.language_model.layers.4.mlp.up_proj	3,670,016	358,852,736
model.language_model.layers.4. post_attention_layernorm	1,024	358,853,760
model.language_model.layers.5.input_layernorm	1,024	358,854,784
model.language_model.layers.5.linear_attn	32	358,854,816
model.language_model.layers.5.linear_attn.conv1d	24,576	358,879,392
model.language_model.layers.5.linear_attn.in_proj_a	16,384	358,895,776
model.language_model.layers.5.linear_attn.in_proj_b	16,384	358,912,160
model.language_model.layers.5.linear_attn. in_proj_qkv	6,291,456	365,203,616
model.language_model.layers.5.linear_attn.in_proj_z	2,097,152	367,300,768
model.language_model.layers.5.linear_attn.norm	128	367,300,896
model.language_model.layers.5.linear_attn.out_proj	2,097,152	369,398,048
model.language_model.layers.5.mlp.down_proj	3,670,016	373,068,064
model.language_model.layers.5.mlp.gate_proj	3,670,016	376,738,080
model.language_model.layers.5.mlp.up_proj	3,670,016	380,408,096
model.language_model.layers.5. post_attention_layernorm	1,024	380,409,120
model.language_model.layers.6.input_layernorm	1,024	380,410,144
model.language_model.layers.6.linear_attn	32	380,410,176
model.language_model.layers.6.linear_attn.conv1d	24,576	380,434,752
model.language_model.layers.6.linear_attn.in_proj_a	16,384	380,451,136
model.language_model.layers.6.linear_attn.in_proj_b	16,384	380,467,520
model.language_model.layers.6.linear_attn. in_proj_qkv	6,291,456	386,758,976
model.language_model.layers.6.linear_attn.in_proj_z	2,097,152	388,856,128
model.language_model.layers.6.linear_attn.norm	128	388,856,256
model.language_model.layers.6.linear_attn.out_proj	2,097,152	390,953,408

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Module	Params	Running total
model.language_model.layers.6.mlp.down_proj	3,670,016	394,623,424
model.language_model.layers.6.mlp.gate_proj	3,670,016	398,293,440
model.language_model.layers.6.mlp.up_proj	3,670,016	401,963,456
model.language_model.layers.6. post_attention_layernorm	1,024	401,964,480
model.language_model.layers.7.input_layernorm	1,024	401,965,504
model.language_model.layers.7.mlp.down_proj	3,670,016	405,635,520
model.language_model.layers.7.mlp.gate_proj	3,670,016	409,305,536
model.language_model.layers.7.mlp.up_proj	3,670,016	412,975,552
model.language_model.layers.7. post_attention_layernorm	1,024	412,976,576
model.language_model.layers.7.self_attn.k_norm	256	412,976,832
model.language_model.layers.7.self_attn.k_proj	524,288	413,501,120
model.language_model.layers.7.self_attn.o_proj	2,097,152	415,598,272
model.language_model.layers.7.self_attn.q_norm	256	415,598,528
model.language_model.layers.7.self_attn.q_proj	4,194,304	419,792,832
model.language_model.layers.7.self_attn.v_proj	524,288	420,317,120
model.language_model.layers.8.input_layernorm	1,024	420,318,144
model.language_model.layers.8.linear_attn	32	420,318,176
model.language_model.layers.8.linear_attn.conv1d	24,576	420,342,752
model.language_model.layers.8.linear_attn.in_proj_a	16,384	420,359,136
model.language_model.layers.8.linear_attn.in_proj_b	16,384	420,375,520
model.language_model.layers.8.linear_attn. in_proj_qkv	6,291,456	426,666,976
model.language_model.layers.8.linear_attn.in_proj_z	2,097,152	428,764,128
model.language_model.layers.8.linear_attn.norm	128	428,764,256
model.language_model.layers.8.linear_attn.out_proj	2,097,152	430,861,408
model.language_model.layers.8.mlp.down_proj	3,670,016	434,531,424
model.language_model.layers.8.mlp.gate_proj	3,670,016	438,201,440
model.language_model.layers.8.mlp.up_proj	3,670,016	441,871,456
model.language_model.layers.8. post_attention_layernorm	1,024	441,872,480
model.language_model.layers.9.input_layernorm	1,024	441,873,504
model.language_model.layers.9.linear_attn	32	441,873,536
model.language_model.layers.9.linear_attn.conv1d	24,576	441,898,112
model.language_model.layers.9.linear_attn.in_proj_a	16,384	441,914,496
model.language_model.layers.9.linear_attn.in_proj_b	16,384	441,930,880
model.language_model.layers.9.linear_attn. in_proj_qkv	6,291,456	448,222,336
model.language_model.layers.9.linear_attn.in_proj_z	2,097,152	450,319,488
model.language_model.layers.9.linear_attn.norm	128	450,319,616
model.language_model.layers.9.linear_attn.out_proj	2,097,152	452,416,768
model.language_model.layers.9.mlp.down_proj	3,670,016	456,086,784
model.language_model.layers.9.mlp.gate_proj	3,670,016	459,756,800
model.language_model.layers.9.mlp.up_proj	3,670,016	463,426,816
model.language_model.layers.9. post_attention_layernorm	1,024	463,427,840
model.language_model.layers.10.input_layernorm	1,024	463,428,864
model.language_model.layers.10.linear_attn	32	463,428,896
model.language_model.layers.10.linear_attn.conv1d	24,576	463,453,472
model.language_model.layers.10.linear_attn. in_proj_a	16,384	463,469,856
model.language_model.layers.10.linear_attn. in_proj_b	16,384	463,486,240
model.language_model.layers.10.linear_attn. in_proj_qkv	6,291,456	469,777,696
model.language_model.layers.10.linear_attn. in_proj_z	2,097,152	471,874,848

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Module	Params	Running total
model.language_model.layers.10.linear_attn.norm	128	471,874,976
model.language_model.layers.10.linear_attn.out_proj	2,097,152	473,972,128
model.language_model.layers.10.mlp.down_proj	3,670,016	477,642,144
model.language_model.layers.10.mlp.gate_proj	3,670,016	481,312,160
model.language_model.layers.10.mlp.up_proj	3,670,016	484,982,176
model.language_model.layers.10.post_attention_layernorm	1,024	484,983,200
model.language_model.layers.11.input_layernorm	1,024	484,984,224
model.language_model.layers.11.mlp.down_proj	3,670,016	488,654,240
model.language_model.layers.11.mlp.gate_proj	3,670,016	492,324,256
model.language_model.layers.11.mlp.up_proj	3,670,016	495,994,272
model.language_model.layers.11.post_attention_layernorm	1,024	495,995,296
model.language_model.layers.11.self_attn.k_norm	256	495,995,552
model.language_model.layers.11.self_attn.k_proj	524,288	496,519,840
model.language_model.layers.11.self_attn.o_proj	2,097,152	498,616,992
model.language_model.layers.11.self_attn.q_norm	256	498,617,248
model.language_model.layers.11.self_attn.q_proj	4,194,304	502,811,552
model.language_model.layers.11.self_attn.v_proj	524,288	503,335,840
model.language_model.layers.12.input_layernorm	1,024	503,336,864
model.language_model.layers.12.linear_attn	32	503,336,896
model.language_model.layers.12.linear_attn.conv1d	24,576	503,361,472
model.language_model.layers.12.linear_attn.in_proj_a	16,384	503,377,856
model.language_model.layers.12.linear_attn.in_proj_b	16,384	503,394,240
model.language_model.layers.12.linear_attn.in_proj_qkv	6,291,456	509,685,696
model.language_model.layers.12.linear_attn.in_proj_z	2,097,152	511,782,848
model.language_model.layers.12.linear_attn.norm	128	511,782,976
model.language_model.layers.12.linear_attn.out_proj	2,097,152	513,880,128
model.language_model.layers.12.mlp.down_proj	3,670,016	517,550,144
model.language_model.layers.12.mlp.gate_proj	3,670,016	521,220,160
model.language_model.layers.12.mlp.up_proj	3,670,016	524,890,176
model.language_model.layers.12.post_attention_layernorm	1,024	524,891,200
model.language_model.layers.13.input_layernorm	1,024	524,892,224
model.language_model.layers.13.linear_attn	32	524,892,256
model.language_model.layers.13.linear_attn.conv1d	24,576	524,916,832
model.language_model.layers.13.linear_attn.in_proj_a	16,384	524,933,216
model.language_model.layers.13.linear_attn.in_proj_b	16,384	524,949,600
model.language_model.layers.13.linear_attn.in_proj_qkv	6,291,456	531,241,056
model.language_model.layers.13.linear_attn.in_proj_z	2,097,152	533,338,208
model.language_model.layers.13.linear_attn.norm	128	533,338,336
model.language_model.layers.13.linear_attn.out_proj	2,097,152	535,435,488
model.language_model.layers.13.mlp.down_proj	3,670,016	539,105,504
model.language_model.layers.13.mlp.gate_proj	3,670,016	542,775,520
model.language_model.layers.13.mlp.up_proj	3,670,016	546,445,536
model.language_model.layers.13.post_attention_layernorm	1,024	546,446,560
model.language_model.layers.14.input_layernorm	1,024	546,447,584
model.language_model.layers.14.linear_attn	32	546,447,616
model.language_model.layers.14.linear_attn.conv1d	24,576	546,472,192

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Module	Params	Running total
model.language_model.layers.14.linear_attn.in_proj_a	16,384	546,488,576
model.language_model.layers.14.linear_attn.in_proj_b	16,384	546,504,960
model.language_model.layers.14.linear_attn.in_proj_qkv	6,291,456	552,796,416
model.language_model.layers.14.linear_attn.in_proj_z	2,097,152	554,893,568
model.language_model.layers.14.linear_attn.norm	128	554,893,696
model.language_model.layers.14.linear_attn.out_proj	2,097,152	556,990,848
model.language_model.layers.14.mlp.down_proj	3,670,016	560,660,864
model.language_model.layers.14.mlp.gate_proj	3,670,016	564,330,880
model.language_model.layers.14.mlp.up_proj	3,670,016	568,000,896
model.language_model.layers.14.post_attention_layernorm	1,024	568,001,920
model.language_model.layers.15.input_layernorm	1,024	568,002,944
model.language_model.layers.15.mlp.down_proj	3,670,016	571,672,960
model.language_model.layers.15.mlp.gate_proj	3,670,016	575,342,976
model.language_model.layers.15.mlp.up_proj	3,670,016	579,012,992
model.language_model.layers.15.post_attention_layernorm	1,024	579,014,016
model.language_model.layers.15.self_attn.k_norm	256	579,014,272
model.language_model.layers.15.self_attn.k_proj	524,288	579,538,560
model.language_model.layers.15.self_attn.o_proj	2,097,152	581,635,712
model.language_model.layers.15.self_attn.q_norm	256	581,635,968
model.language_model.layers.15.self_attn.q_proj	4,194,304	585,830,272
model.language_model.layers.15.self_attn.v_proj	524,288	586,354,560
model.language_model.layers.16.input_layernorm	1,024	586,355,584
model.language_model.layers.16.linear_attn	32	586,355,616
model.language_model.layers.16.linear_attn.conv1d	24,576	586,380,192
model.language_model.layers.16.linear_attn.in_proj_a	16,384	586,396,576
model.language_model.layers.16.linear_attn.in_proj_b	16,384	586,412,960
model.language_model.layers.16.linear_attn.in_proj_qkv	6,291,456	592,704,416
model.language_model.layers.16.linear_attn.in_proj_z	2,097,152	594,801,568
model.language_model.layers.16.linear_attn.norm	128	594,801,696
model.language_model.layers.16.linear_attn.out_proj	2,097,152	596,898,848
model.language_model.layers.16.mlp.down_proj	3,670,016	600,568,864
model.language_model.layers.16.mlp.gate_proj	3,670,016	604,238,880
model.language_model.layers.16.mlp.up_proj	3,670,016	607,908,896
model.language_model.layers.16.post_attention_layernorm	1,024	607,909,920
model.language_model.layers.17.input_layernorm	1,024	607,910,944
model.language_model.layers.17.linear_attn	32	607,910,976
model.language_model.layers.17.linear_attn.conv1d	24,576	607,935,552
model.language_model.layers.17.linear_attn.in_proj_a	16,384	607,951,936
model.language_model.layers.17.linear_attn.in_proj_b	16,384	607,968,320
model.language_model.layers.17.linear_attn.in_proj_qkv	6,291,456	614,259,776
model.language_model.layers.17.linear_attn.in_proj_z	2,097,152	616,356,928
model.language_model.layers.17.linear_attn.norm	128	616,357,056
model.language_model.layers.17.linear_attn.out_proj	2,097,152	618,454,208

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Module	Params	Running total
model.language_model.layers.17.mlp.down_proj	3,670,016	622,124,224
model.language_model.layers.17.mlp.gate_proj	3,670,016	625,794,240
model.language_model.layers.17.mlp.up_proj	3,670,016	629,464,256
model.language_model.layers.17. post_attention_layernorm	1,024	629,465,280
model.language_model.layers.18.input_layernorm	1,024	629,466,304
model.language_model.layers.18.linear_attn	32	629,466,336
model.language_model.layers.18.linear_attn.conv1d	24,576	629,490,912
model.language_model.layers.18.linear_attn. in_proj_a	16,384	629,507,296
model.language_model.layers.18.linear_attn. in_proj_b	16,384	629,523,680
model.language_model.layers.18.linear_attn. in_proj_qkv	6,291,456	635,815,136
model.language_model.layers.18.linear_attn. in_proj_z	2,097,152	637,912,288
model.language_model.layers.18.linear_attn.norm	128	637,912,416
model.language_model.layers.18.linear_attn.out_proj	2,097,152	640,009,568
model.language_model.layers.18.mlp.down_proj	3,670,016	643,679,584
model.language_model.layers.18.mlp.gate_proj	3,670,016	647,349,600
model.language_model.layers.18.mlp.up_proj	3,670,016	651,019,616
model.language_model.layers.18. post_attention_layernorm	1,024	651,020,640
model.language_model.layers.19.input_layernorm	1,024	651,021,664
model.language_model.layers.19.mlp.down_proj	3,670,016	654,691,680
model.language_model.layers.19.mlp.gate_proj	3,670,016	658,361,696
model.language_model.layers.19.mlp.up_proj	3,670,016	662,031,712
model.language_model.layers.19. post_attention_layernorm	1,024	662,032,736
model.language_model.layers.19.self_attn.k_norm	256	662,032,992
model.language_model.layers.19.self_attn.k_proj	524,288	662,557,280
model.language_model.layers.19.self_attn.o_proj	2,097,152	664,654,432
model.language_model.layers.19.self_attn.q_norm	256	664,654,688
model.language_model.layers.19.self_attn.q_proj	4,194,304	668,848,992
model.language_model.layers.19.self_attn.v_proj	524,288	669,373,280
model.language_model.layers.20.input_layernorm	1,024	669,374,304
model.language_model.layers.20.linear_attn	32	669,374,336
model.language_model.layers.20.linear_attn.conv1d	24,576	669,398,912
model.language_model.layers.20.linear_attn. in_proj_a	16,384	669,415,296
model.language_model.layers.20.linear_attn. in_proj_b	16,384	669,431,680
model.language_model.layers.20.linear_attn. in_proj_qkv	6,291,456	675,723,136
model.language_model.layers.20.linear_attn. in_proj_z	2,097,152	677,820,288
model.language_model.layers.20.linear_attn.norm	128	677,820,416
model.language_model.layers.20.linear_attn.out_proj	2,097,152	679,917,568
model.language_model.layers.20.mlp.down_proj	3,670,016	683,587,584
model.language_model.layers.20.mlp.gate_proj	3,670,016	687,257,600
model.language_model.layers.20.mlp.up_proj	3,670,016	690,927,616
model.language_model.layers.20. post_attention_layernorm	1,024	690,928,640
model.language_model.layers.21.input_layernorm	1,024	690,929,664
model.language_model.layers.21.linear_attn	32	690,929,696
model.language_model.layers.21.linear_attn.conv1d	24,576	690,954,272
model.language_model.layers.21.linear_attn. in_proj_a	16,384	690,970,656

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Module	Params	Running total
model.language_model.layers.21.linear_attn.in_proj_b	16,384	690,987,040
model.language_model.layers.21.linear_attn.in_proj_qkv	6,291,456	697,278,496
model.language_model.layers.21.linear_attn.in_proj_z	2,097,152	699,375,648
model.language_model.layers.21.linear_attn.norm	128	699,375,776
model.language_model.layers.21.linear_attn.out_proj	2,097,152	701,472,928
model.language_model.layers.21.mlp.down_proj	3,670,016	705,142,944
model.language_model.layers.21.mlp.gate_proj	3,670,016	708,812,960
model.language_model.layers.21.mlp.up_proj	3,670,016	712,482,976
model.language_model.layers.21.post_attention_layernorm	1,024	712,484,000
model.language_model.layers.22.input_layernorm	1,024	712,485,024
model.language_model.layers.22.linear_attn	32	712,485,056
model.language_model.layers.22.linear_attn.conv1d	24,576	712,509,632
model.language_model.layers.22.linear_attn.in_proj_a	16,384	712,526,016
model.language_model.layers.22.linear_attn.in_proj_b	16,384	712,542,400
model.language_model.layers.22.linear_attn.in_proj_qkv	6,291,456	718,833,856
model.language_model.layers.22.linear_attn.in_proj_z	2,097,152	720,931,008
model.language_model.layers.22.linear_attn.norm	128	720,931,136
model.language_model.layers.22.linear_attn.out_proj	2,097,152	723,028,288
model.language_model.layers.22.mlp.down_proj	3,670,016	726,698,304
model.language_model.layers.22.mlp.gate_proj	3,670,016	730,368,320
model.language_model.layers.22.mlp.up_proj	3,670,016	734,038,336
model.language_model.layers.22.post_attention_layernorm	1,024	734,039,360
model.language_model.layers.23.input_layernorm	1,024	734,040,384
model.language_model.layers.23.mlp.down_proj	3,670,016	737,710,400
model.language_model.layers.23.mlp.gate_proj	3,670,016	741,380,416
model.language_model.layers.23.mlp.up_proj	3,670,016	745,050,432
model.language_model.layers.23.post_attention_layernorm	1,024	745,051,456
model.language_model.layers.23.self_attn.k_norm	256	745,051,712
model.language_model.layers.23.self_attn.k_proj	524,288	745,576,000
model.language_model.layers.23.self_attn.o_proj	2,097,152	747,673,152
model.language_model.layers.23.self_attn.q_norm	256	747,673,408
model.language_model.layers.23.self_attn.q_proj	4,194,304	751,867,712
model.language_model.layers.23.self_attn.v_proj	524,288	752,392,000
model.language_model.norm	1,024	752,393,024
model.visual.blocks.0.attn.proj	590,592	752,983,616
model.visual.blocks.0.attn.qkv	1,771,776	754,755,392
model.visual.blocks.0.mlp.linear_fc1	2,362,368	757,117,760
model.visual.blocks.0.mlp.linear_fc2	2,360,064	759,477,824
model.visual.blocks.0.norm1	1,536	759,479,360
model.visual.blocks.0.norm2	1,536	759,480,896
model.visual.blocks.1.attn.proj	590,592	760,071,488
model.visual.blocks.1.attn.qkv	1,771,776	761,843,264
model.visual.blocks.1.mlp.linear_fc1	2,362,368	764,205,632
model.visual.blocks.1.mlp.linear_fc2	2,360,064	766,565,696
model.visual.blocks.1.norm1	1,536	766,567,232
model.visual.blocks.1.norm2	1,536	766,568,768
model.visual.blocks.2.attn.proj	590,592	767,159,360
model.visual.blocks.2.attn.qkv	1,771,776	768,931,136

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Module	Params	Running total
model.visual.blocks.2.mlp.linear_fc1	2,362,368	771,293,504
model.visual.blocks.2.mlp.linear_fc2	2,360,064	773,653,568
model.visual.blocks.2.norm1	1,536	773,655,104
model.visual.blocks.2.norm2	1,536	773,656,640
model.visual.blocks.3.attn.proj	590,592	774,247,232
model.visual.blocks.3.attn.qkv	1,771,776	776,019,008
model.visual.blocks.3.mlp.linear_fc1	2,362,368	778,381,376
model.visual.blocks.3.mlp.linear_fc2	2,360,064	780,741,440
model.visual.blocks.3.norm1	1,536	780,742,976
model.visual.blocks.3.norm2	1,536	780,744,512
model.visual.blocks.4.attn.proj	590,592	781,335,104
model.visual.blocks.4.attn.qkv	1,771,776	783,106,880
model.visual.blocks.4.mlp.linear_fc1	2,362,368	785,469,248
model.visual.blocks.4.mlp.linear_fc2	2,360,064	787,829,312
model.visual.blocks.4.norm1	1,536	787,830,848
model.visual.blocks.4.norm2	1,536	787,832,384
model.visual.blocks.5.attn.proj	590,592	788,422,976
model.visual.blocks.5.attn.qkv	1,771,776	790,194,752
model.visual.blocks.5.mlp.linear_fc1	2,362,368	792,557,120
model.visual.blocks.5.mlp.linear_fc2	2,360,064	794,917,184
model.visual.blocks.5.norm1	1,536	794,918,720
model.visual.blocks.5.norm2	1,536	794,920,256
model.visual.blocks.6.attn.proj	590,592	795,510,848
model.visual.blocks.6.attn.qkv	1,771,776	797,282,624
model.visual.blocks.6.mlp.linear_fc1	2,362,368	799,644,992
model.visual.blocks.6.mlp.linear_fc2	2,360,064	802,005,056
model.visual.blocks.6.norm1	1,536	802,006,592
model.visual.blocks.6.norm2	1,536	802,008,128
model.visual.blocks.7.attn.proj	590,592	802,598,720
model.visual.blocks.7.attn.qkv	1,771,776	804,370,496
model.visual.blocks.7.mlp.linear_fc1	2,362,368	806,732,864
model.visual.blocks.7.mlp.linear_fc2	2,360,064	809,092,928
model.visual.blocks.7.norm1	1,536	809,094,464
model.visual.blocks.7.norm2	1,536	809,096,000
model.visual.blocks.8.attn.proj	590,592	809,686,592
model.visual.blocks.8.attn.qkv	1,771,776	811,458,368
model.visual.blocks.8.mlp.linear_fc1	2,362,368	813,820,736
model.visual.blocks.8.mlp.linear_fc2	2,360,064	816,180,800
model.visual.blocks.8.norm1	1,536	816,182,336
model.visual.blocks.8.norm2	1,536	816,183,872
model.visual.blocks.9.attn.proj	590,592	816,774,464
model.visual.blocks.9.attn.qkv	1,771,776	818,546,240
model.visual.blocks.9.mlp.linear_fc1	2,362,368	820,908,608
model.visual.blocks.9.mlp.linear_fc2	2,360,064	823,268,672
model.visual.blocks.9.norm1	1,536	823,270,208
model.visual.blocks.9.norm2	1,536	823,271,744
model.visual.blocks.10.attn.proj	590,592	823,862,336
model.visual.blocks.10.attn.qkv	1,771,776	825,634,112
model.visual.blocks.10.mlp.linear_fc1	2,362,368	827,996,480
model.visual.blocks.10.mlp.linear_fc2	2,360,064	830,356,544
model.visual.blocks.10.norm1	1,536	830,358,080
model.visual.blocks.10.norm2	1,536	830,359,616
model.visual.blocks.11.attn.proj	590,592	830,950,208
model.visual.blocks.11.attn.qkv	1,771,776	832,721,984
model.visual.blocks.11.mlp.linear_fc1	2,362,368	835,084,352
model.visual.blocks.11.mlp.linear_fc2	2,360,064	837,444,416
model.visual.blocks.11.norm1	1,536	837,445,952
model.visual.blocks.11.norm2	1,536	837,447,488

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Module	Params	Running total
model.visual.merger.linear_fc1	9,440,256	846,887,744
model.visual.merger.linear_fc2	3,146,752	850,034,496
model.visual.merger.norm	1,536	850,036,032
model.visual.patch_embed.proj	1,180,416	851,216,448
model.visual.pos_embed	1,769,472	852,985,920